

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

Step 1 : Calculate the quantity of heat required to convert a liter of H₂O at 30°C to a liter of water at 60°C .

To calculate the heat required , we use the formula $Q = mc\Delta T$, where m is the mass of the substance , c is the specific heat capacity , and ΔT is the change in temperature .

For water , $m = 1\text{ L} = 1000\text{ g}$ (since the density of water is approximately 1 g/mL), $c = 1\text{ cal/g}^\circ\text{C}$, and $\Delta T = 60^\circ\text{C} - 30^\circ\text{C} = 30^\circ\text{C}$.

$$Q = mc\Delta T = 1000\text{ g} * 1\text{ cal/g}^\circ\text{C} * 30^\circ\text{C} = 30,000\text{ cal} = 30\text{ Kcal} .$$

Step 2 : Calculate the quantity of heat required to heat a 1 kg block of aluminum from 30° to 60°C .

For aluminum , $m = 1\text{ kg} = 1000\text{ g}$, $c = 0.215\text{ cal/g}^\circ\text{C}$, and $\Delta T = 60^\circ\text{C} - 30^\circ\text{C} = 30^\circ\text{C}$.

$$Q = mc\Delta T = 1000\text{ g} * 0.215\text{ cal/g}^\circ\text{C} * 30^\circ\text{C} = 6425\text{ cal} = 6.425\text{ Kcal} \approx 6.45\text{ Kcal} .$$

Therefore , the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C is 30 Kcal , and (b) heat a 1 kg block of aluminum from 30° to 60°C is 6.45 Kcal .

The correct answer is G . 30 Kcal , 6.45 Kcal .

</think><answer> G . 30 Kcal , 6.45 Kcal </answer>